

COS80023 Big Data – Lab 5

Lab 5: Pass Task 5 – Information Retrieval with Pig

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Question 1: What does each part of this line do?

```
rawsales = LOAD '/pigdata/SalesJan2009.csv' USING PigStorage(',')
AS (Transaction_date:chararray, Product:chararray, Price:int,
Payment_type:chararray, Name:chararray, City:chararray,
State:chararray, Country:chararray);
```

- `rawsales = LOAD '/pigdata/SalesJan2009.csv'` → Loads the CSV file from the directory /pigdata into Pig.
- `USING PigStorage(',')` → Tells Pig to use a comma as the field separator.
- `AS (...)` → Defines the schema (column names and data types).
 - `Transaction_date:chararray` → stores the date as a string.
 - `Product:chararray` → stores the product name.
 - `Price:int` → stores the price as an integer.
 - `Payment_type:chararray` → type of payment (e.g., Visa, MasterCard).
 - `Name, City, State, Country` → customer details stored as strings.

This command loads the raw CSV file and maps it into a structured dataset that Pig can process.

```
DUMP rawsales;
```

- It prints the contents of the relation `rawsales` to the screen.

```
salesdata = FILTER rawsales BY Price IS NOT NULL;
```

- Removes the header row and any invalid rows where the Price field is missing.
- Ensures only meaningful sales data is kept.

Question 2: What does each line do (explain every detail)?

```
salesandcountry = FOREACH salesdata GENERATE (chararray) Country AS
country, (int) Price AS price;
```

- Creates a new relation with only two columns: country and price.
- Explicitly casts Country as a string and Price as an integer.

```
countrygroups = GROUP salesandcountry BY country;
```

- Groups all sales transactions by country.
- Prepares the dataset for aggregation (summing).

```
salespercountry = FOREACH countrygroups GENERATE group as country,
SUM(salesandcountry.price) AS totalsales;
```

- For each country, calculates the **total sales amount** by summing all prices in that group.
- Outputs a dataset with country and totalsales.

Question 3: What do these two lines do?

```
sortedsales = ORDER salespercountry BY totalsales desc;
```

- Sorts the countries in descending order of total sales.
- Ensures the highest sales appear first.

```
STORE sortedsales INTO '/pigdata/SalesByCountry' USING PigStorage(' ');
```

- Saves the sorted results into a new directory called SalesByCountry inside /pigdata.
- Uses space (' ') as the field separator in the output file.

Question 4: Document your code for sales per credit card.

To calculate sales per credit card type, I modified the Pig script as follows:

```
salesandpayment = FOREACH salesdata GENERATE (chararray)
Payment_type AS payment, (int) Price AS price;
```

```
paymentgroups = GROUP salesandpayment BY payment;
```

```
salesperpayment = FOREACH paymentgroups GENERATE group as
payment_type, SUM(salesandpayment.price) AS totalsales;
```

```
sortedsalespayment = ORDER salesperpayment BY totalsales desc;
```

```
STORE sortedsalespayment INTO '/pigdata/SalesByPaymentType' USING
PigStorage(' ');
```

- Extracts only Payment_type and Price.
- Groups by Payment_type (e.g., Visa, MasterCard, Amex).
- Sums the sales per payment type.
- Orders the results by total sales in descending order.
- Stores the results in /pigdata/SalesByPaymentType.

```
sshuser@hn0-s10483: ~
grunt> dump sortedsalespayment
```

```
2025-09-04 05:45:56,564 [main] INFO org.apache.hadoop.mapreduce.lib.in
put.FileInputFormat - Total input files to process : 1
2025-09-04 05:45:56,564 [main] INFO org.apache.pig.backend.hadoop.exec
utionengine.util.MapRedUtil - Total input paths to process : 1
(Visa,836350)
(Mastercard,458450)
(Amex,188900)
(Diners,133800)
grunt>
```

DUMP of sorted credit card payments

Question 5: Repeat the wordcount example in Pig.

Steps followed:

1. I Uploaded a text file to /pigdata/wordinput.
2. I Ran the following Pig Latin commands:

```
lines = LOAD '/pigdata/wordinput' USING PigStorage('\n') AS
(line:chararray);
```

```
words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) AS word;
```

```
wordgroups = GROUP words BY word;
```

```
wordcount = FOREACH wordgroups GENERATE group AS word, COUNT(words)
AS count;
```

```
STORE wordcount INTO '/pigdata/WordCountOutput' USING PigStorage('
');
```

- LOAD → loads the text file line by line.

- TOKENIZE → splits each line into words.
- FLATTEN → converts tokenized words into individual rows.
- GROUP → groups the words by their value.
- COUNT → counts how many times each word appears.
- STORE → writes the results to /pigdata/WordCountOutput.

DUMP wordcount;

```
sshuser@hn0-s10483: ~
(,0)
grunt> dump wordcount;
```

```
(Evaluation,1)
(Inferences,1)
(Learnings:,17)
(Microsoft.,1)
(Moderately,1)
(Refinement,1)
(Robustness,1)
(WalledEval,1)
(accessible,1)
(accounting,1)
(alignment.,1)
(attractive,1)
(benchmarks,4)
(categories,5)
(classified,1)
(compliance,1)
(condition:,3)
(considered,1)
(dangerous:,1)
(effective.,3)
(efficiency,1)
(encourages,1)
(escalation,1)
(especially,2)
(evaluation,8)
(exploited.,1)
(fine-tuned,2)
(generating,1)
(government,1)
```

DUMP of Wordcount from the text file

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